

Literature Benchmark Brief: Karpatne_2017_TKDE

Title: Theory-guided data science: Capabilities and challenges

Authors: A. Karpatne, G. Atluri, P. R. Faghmous, M. Steinbach, A. Banerjee, A. Ganguly, S. Shekhar, N. Samatova, V. Kumar (2017)

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Cluster: Cluster E: Missing Data & Modular Risk Layers

Core Theoretical Mechanism & Model Formulation

Integrating physical principles into data science: physics-guided design and loss constraints

Empirical Scale & Experimental Dataset

Climate science, lake temperature modeling, materials engineering

Relevance to IEEE TSTE & Sustainable Energy Systems

Foundational IEEE paper articulating the theory-guided machine learning paradigm

Identified Gaps & Operational Vulnerabilities

High-level conceptual overview without domain-specific wind turbine power formulas

Role in Manuscript Defense & Reviewer #2 Proof

Directly cited in Section II for theory-guided machine learning formulation